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Sounding Rocket Geophysical Environment 10 February 2010

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Introduction

Accurate knowledge of the geophysical environment is essential for a great many sounding rocket analysis tasks. Meeting this need means more than a simple recapitulation of the usual textbook results. Sounding rockets need a more refined atmosphere model than the usual standard, but a less detailed geodesy and gravity descriptions.

The model described in this memo is realized in GEOPHYSICS.xls Note that the cells with a sky blue fill require user inputs.

Atmosphere Model

Sounding rockets have flight dynamics reminiscent of light aircraft. Due to their small size, they are sensitive to low altitude density altitude effects. This implies that we should use a local temperature-altitude profile reflecting the actual surface temperature and troposphere height. Since we seek a local atmosphere model, the other parameters in the Standard Atmosphere are retained.

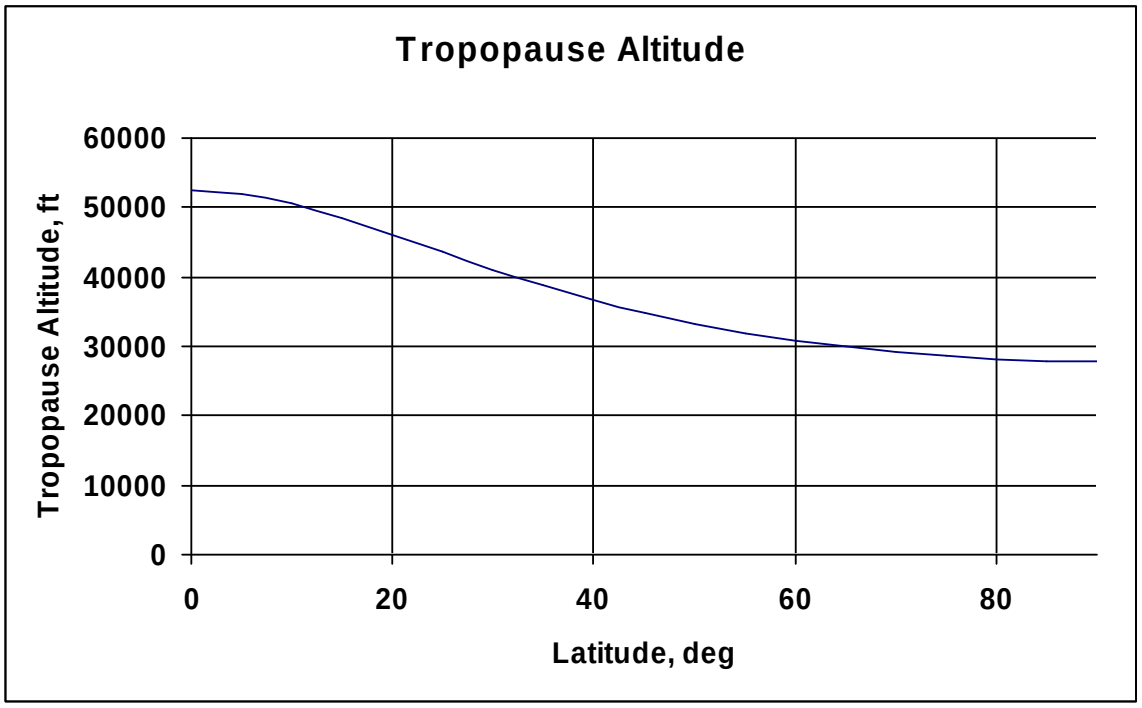
The absolute temperature profile in the atmosphere is modeled by a sequence of straight line elements, each covering an atmospheric layer. For most university sounding rockets, we need only consider the lowest two layers, the troposphere and the stratosphere. Their boundary is called the tropopause. The troposphere is characterized by weather and the stratosphere by its greater stability.

It has long been known that the tropopause is far lower in the arctic than near the equator. We model the tropopause with an ellipsoid

$$\mathbf{h_T^2 = 1 / [(\cos\varphi / a)^2 + (\sin\varphi / b)^2]},$$

where $\mathbf{h_T}$ = Altitude of the tropopause above the geoid,
$\boldsymbol{\varphi}$ = Geodetic latitude,
$\mathbf{a}$ = Altitude of the equatorial tropopause ≈ 52,500 ft, and
$\mathbf{b}$ = Altitude of the polar tropopause ≈ 27,900 ft.

The numbers above are from ref. (1).



The tropospheric temperature is dominated by convective mixing. Parcels of air near the surface are warmed by contact with the hot ground, break free and ascend through the atmosphere just like a hot air balloon. As a parcel rises, it expands and cools adiabatically (without any external heat transfer). These parcels, called thermals, are the source of atmospheric turbulence and bumpy airplane rides. Condensation of water vapor modifies parcel cooling so that the average temperature lapse rate (**dT/dh**) is only about 75-80% of that for an ideal thermal expanding adiabatically. Note that there is very little convective mixing and turbulence above the tropopause leading to an isothermal stratosphere.

Now, suppose the temperature **T*** is specified at a altitude **h***. For example, this could be the temperature at a launch site. Now the temperature decreases linearly in the troposphere so the tropopause temperature $\mathbf{T_T}$ is just

$$\mathbf{T_T} = \mathbf{T^*} + \alpha\,(\mathbf{h_T} - \mathbf{h^*}),$$

where α = Temperature lapse rate = -0.00357448 ^{o}R/ft.

The lapse rate is also taken from ref. (1).

The temperature above the tropopause, the stratospheric temperature, is invariant with altitude. Note that the lower polar tropopause implies the polar stratosphere is warmer than that in the tropics. This counterintuitive result is well documented in ref. (1).

The rest of the atmosphere model is based on the perfect gas law and hydrostatic equilibrium. The perfect gas law states

$$\mathbf{p = \rho R T},$$

where $\mathbf{p}$ = Atmospheric pressure, lb/ft^2,
ρ = Atmospheric mass density, sl/ft^3,
$\mathbf{R}$ = Gas constant for air = 1716 ^{o}R/ft, and
$\mathbf{T}$ = Atmosphere absolute temperature, ^{o}R.
Note the gas constant for air is based on a mean molecular weight of 28.9644 per ref. (1).

The condition of hydrostatic equilibrium is

$$\mathbf{dp/dh = -\rho g},$$

where $\mathbf{g}$ = local acceleration of gravity (See the following section)

After a modest amount of calculus, the pressure as a function of altitude $\mathbf{h}$ is found to be

$$\mathbf{p = p_{SL} \left(1 / (1 + \alpha h / (T^* - \alpha h^*))\right)^{(-g / R \alpha)}}, \text{ (troposphere), and}$$

$$\mathbf{p = p_T \exp(-g (h - h_T) / R T_T)} \text{ (stratosphere, } \alpha = 0)$$

Here, $\mathbf{p_{SL}}$ = Sea level (geoid) pressure = 2116.217 lb/ft^2, and
$\mathbf{p_T}$ = Tropopause pressure = $\mathbf{p_{SL} \left(1 / (1 + \alpha h_T / (T^* - \alpha h^*))\right)^{(-g / R \alpha)}}$.

The remaining state variables can easily be found. Density comes from the perfect gas law knowing pressure and temperature. Sound speed is given by

$$\mathbf{a^2 = \gamma R T},$$

where γ = ratio of specific heats for air cp/cv = 1.400.

Geodesy and Gravity

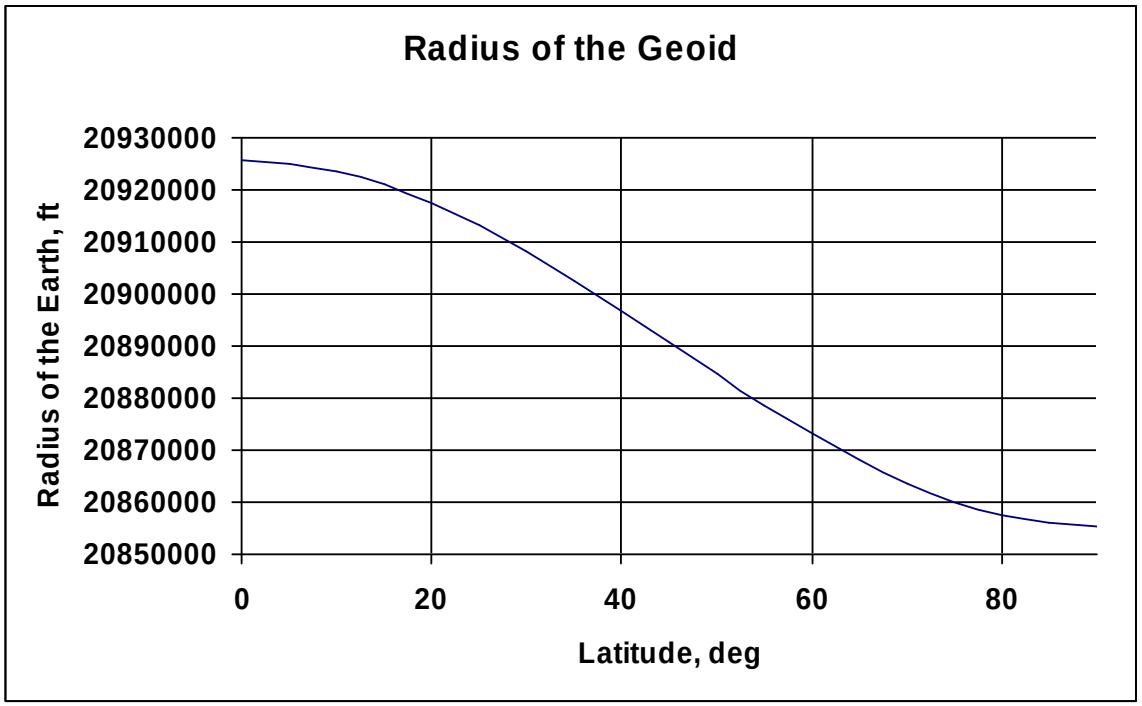
Geodesy is the science of the Earth's shape (the geoid); gravity is the science of its attractive force. To describe the shape of the Earth, first identify two distinct latitudes. These are geocentric latitude, defined as the angle between the equatorial plane and a radius vector from the center of the Earth, and geodetic latitude, defined as the angle between the equatorial plane and a vector normal to the geoid. Because the Earth spins

its shape is slightly flattened at its poles. Most maps and trajectory codes use geodetic latitude.
Model the Earth's shape as a sphere tangent to the geoid at a specified latitude. Model the geoid as an ellipsoid of revolution whose radius $\mathbf{R_E}$ is given by

$$\mathbf{R_E^2 = ((a^2 \cos\varphi)^2 + (b^2 \sin\varphi)^2) / ((a \cos\varphi)^2 + (b \sin\varphi)^2)}.$$

Here **a =** Equatorial radius of the Earth = 20,925,597.9 ft, and
b = Polar radius of the Earth = 20,855,437.3 ft.

Both radii are consisted with WGS 84 as reported in ref. (2). Thus, knowing the geodetic latitude, one finds the geodetic radius of the tangent sphere from the result above.
Finally, the Earth's rotation rate is given by 2π / 86164.09 rad/sec. 86164.09 sec is the Earth's rotation period as observed against the remote stars.



The facts that on the geoid centripetal acceleration varies with latitude as does the distance from the center of the Earth guarantee that surface gravity will also vary with latitude. This is captured in another result from ref. (2), the Potsdam result:

$$\mathbf{g(\varphi,0) = 32.0876228*(1 + 0.00530224 \sin^2(\varphi) - 0.000058*\sin^2(2\varphi))},$$

where **g(φ,0) =** The acceleration of gravity at latitude **φ** and altitude 0. Note that **g** is in ft/sec^2. The so-called "standard g" = 32.174 ft/sec^2 used for a great deal of engineering work is just a mid-latitude evaluation of this formula.

The approximate inverse square correction for a non-zero altitude **h** is just

$$\mathbf{g(\varphi,h) = g(\varphi,0)*(R_E / (R_E + h))^2}.$$

References

1. COESA, "U.S. Standard Atmosphere, 1976", U.S. Government Printing Office, Washington, D. C., 1976
2. Air Force Geophysics Laboratory, "Handbook of Geophysics and the Space Environment", Third Edition, Air Force Geophysics Laboratory, 1985.